

BIOLOGY

Biology looks at how animals, plants, and other living things work. Learning about the human body allows scientists to develop crucial medicine, and understanding how trees grow can help us fight climate change.

THE SCIENTIFIC METHOD

There are many different areas of science, but most scientists follow a method in their quest to understand the world. The most important part of the scientific method is experimentation, which allows scientists to test their hypotheses (suggested explanations).

Observation

The scientific method begins with careful observation of some aspect of the world, something no one can yet fully explain. Scientists may observe something in a laboratory or in the wider world around us.

Hypothesis

Next, a scientist comes up with a possible explanation for their observation: a hypothesis. The hypothesis should be based on existing scientific knowledge and must be something that can be tested by carrying out an experiment.

Experiment

A scientist must design their experiments carefully, making sure they control as many aspects of the setup as possible. Typically, they will predict an outcome of the experiment that, if true, will support or disprove their hypothesis.

Analysis and theory

If a hypothesis is supported by the outcome of the experiments, that hypothesis may become part of a theory—a generally accepted part of science that may explain lots of different observations. Well-tested theories include the theory of evolution and the Big Bang theory.

CHEMISTRY

Chemistry is the study of how different substances react together. Using their knowledge, chemists can create new materials, such as synthetic textiles, and can also develop ways to extract key resources, such as salt.